

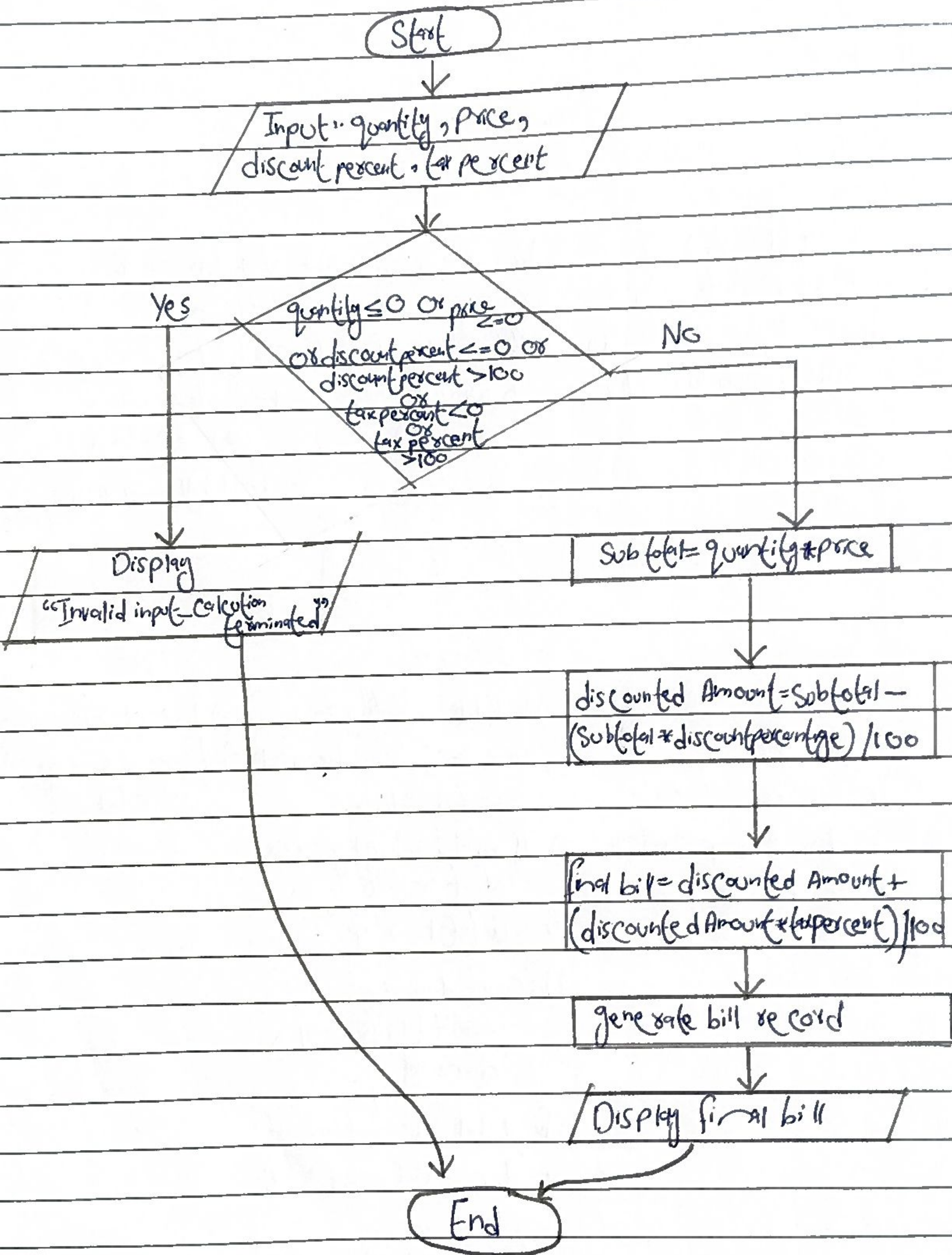
Question 4

PAC

Input Data	Output	Processing Required
quantity (q) Price per item (p) discount percentage (d) tax percentage (t)	Final bill amount or an error message if input is valid	Validate all four inputs $\text{Subtotal (S)} = q \times p$ discounted amount $(a) = S - (S \times d) / 100$ final bill = $a + (a \times t) / 100$

IPO

Input	Process	Output
- quantity - price per item - discount percentage - tax percentage	- Validate quantity > 0, price > 0, $0 \leq \text{discount} \leq 100$, $0 \leq \text{tax} \leq 100$ - If invalid $\rightarrow$ display error and terminate - Subtotal = $q \times p$ - discounted amount = $\text{Subtotal} - (\text{Subtotal} \times \text{discount}) / 100$ - final bill = discounted amount + $(\text{discounted amount} \times \text{tax}) / 100$ - Generate bill records	- Final bill amount - Error message if invalid



Pseudocode

Begin

Input quantity, price, discount percent, tax percent

If

quantity ≤ 0 or price ≤ 0 or discount percent ≤ 0 or
discount percent ≥ 100 or tax percent ≤ 0 or tax percent
 ≥ 100

Display "Error - Invalid Input. Calculation terminated"

Else

Subtotal = quantity * price

discounted Amount = Subtotal - (Subtotal * discount percentage) / 100

final bill = discounted amount + (discounted amount * tax percent) / 100

Store

quantity, price, discount percent, tax percent, final bill as
bill record

Display "Subtotal", Subtotal

Display "Discounted Amount", discounted Amount

Display "Final bill", final bill

End If

End